

DATABASE SCHEMA • V1.0

Backend Schema & API Contracts

MongoDB Document Models, Index Architecture & Complete
REST API Catalog

PROJECT

Gymholik SaaS

VERSION

1.0.0

AUTHOR

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DATE

October 2026

1. MongoDB Document Schemas

Every collection is partitioned by gymId to ensure multi-tenant security. Compound indexes ensure sub-millisecond query execution.

1.1 Collection: gyms

Field	Type	Index	Description
_id	ObjectId	Primary	Unique gym tenant ID.
gymName	String	Text	Registered gym title.
gymCode	String	Unique	Collision-free uppercase 6-10 char code (e.g. TITANF4067).
ownerId	ObjectId	Indexed	Foreign key to users table for owner.
trialEndsAt	ISODate	Sparse	Trial termination timestamp.
subscriptionActive	Boolean	Indexed	Subscription status flag.
settings	Embedded	-	allowExpiredCheckin, maskPhone, reminderDays.

1.2 Collection: users

Field	Type	Index	Description
_id	ObjectId	Primary	User identifier.
gymId	ObjectId	Compound (gymId, phone)	Tenant association.
phone	String	Indexed	Unique credential for login.
passwordHash	String	-	BCrypt salted password hash.
name	String	-	Full name of user.
role	String	Indexed	OWNER, STAFF, TRAINER, SUPER_ADMIN.
status	String	Indexed	ACTIVE, PENDING_APPROVAL, SUSPENDED.
trainerTypeIds	Array[ObjectId]	-	Assigned specialization tags for trainers.

1.3 Collection: members

Field	Type	Index	Description
_id	ObjectId	Primary	Member ID.
gymId	ObjectId	Compound (gymId, status)	Tenant partition.
memberCode	String	Compound (gymId, memberCode)	Auto-incremented gym member code.
name	String	Compound (gymId, name)	Full member name.
phone	String	Compound (gymId, phone)	Member contact mobile.
currentExpiry	ISODate	Indexed	Active membership end date.
currentDuePaise	Long	Indexed	Pending dues in paise (1 INR = 100 paise).
status	String	Indexed	ACTIVE, EXPIRED.

1.4 Collection: payments

Field	Type	Index	Description
_id	ObjectId	Primary	Payment record identifier.
gymId	ObjectId	Compound (gymId, createdAt)	Tenant isolation.
memberId	ObjectId	Indexed	Target member foreign key.
amountPaise	Long	-	Transacted sum in paise.
mode	String	Indexed	CASH, UPI, CARD, BANK_TRANSFER.
type	String	Indexed	PAYMENT, REVERSAL.
collectedBy	ObjectId	-	User ID of staff/owner recording entry.

1.5 Collection: attendance

Field	Type	Index	Description
_id	ObjectId	Primary	Check-in log ID.
gymId	ObjectId	Compound (gymId, checkInDate)	Tenant isolation.
memberId	ObjectId	Indexed	Checked-in member ID.
checkInTime	ISODate	Indexed	Exact timestamp of access.
checkInDate	String (YYYY-MM-DD)	Compound	Date bucket for daily aggregations.
method	String	-	MANUAL, QR_SCAN.

2. Complete REST API Catalog

Endpoint	Method	Auth & Role	Description
/api/auth/register-gym	POST	Public	Registers tenant and starts 14-day trial.
/api/auth/register-trainer	POST	Public	Registers trainer in PENDING_APPROVAL status.
/api/auth/login	POST	Public	Returns access + refresh tokens.
/api/auth/me	GET	Authenticated	Returns authenticated user identity & profile.
/api/dashboard	GET	OWNER, STAFF	Returns financial metrics & 7-day collections.
/api/members	GET	Authenticated	Paged member list (trainers scoped to assigned).
/api/members	POST	OWNER, STAFF	Enrolls new member with plan & payment.
/api/attendance/check-in	POST	Authenticated	Records member check-in timestamp.
/api/payments	GET	OWNER, STAFF	Paginated payment ledger with date/mode filters.
/api/payments/{id}/reverse	POST	OWNER Only	Reverses payment transaction with audit note.
/api/trainers/{id}/approve	POST	OWNER Only	Approves trainer and binds trainer types.